

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Date of Assignment : 15/07/2026	Date/Time of Submission : 15/07/2026

Queue Application:

- a. Write a program to: a. Implement a queue using an array.

CODE :

```
import array as arr

que = arr.array('i', [])

si = 4

print(que)

def enqueue(data):
    if(len(que) == si):
        print("Queue Overflow")
    else:
        que.append(data)
        print(que)

def dequeue():
    if(len(que) == 0):
        print("Queue Underflow")
    else:
        que.pop(0)
        print("Data removed: \n", que)

def peek():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("First element is: ", que[0])

def show():
    if(len(que) == 0):
        print("Queue is Empty")
    else:
        print("---- QUEUE ---- \n", que)

enqueue(2)
dequeue()
dequeue()
enqueue(3)
enqueue(4)
enqueue(5)
enqueue(6)
enqueue(7)

dequeue()
peek()
dequeue()
show()
print("\nAbhishek S025")
```

OUTPUT :

```
array('i')
array('i', [2])
Data removed:
array('i')
Queue Underflow
array('i', [3])
array('i', [3, 4])
array('i', [3, 4, 5])
array('i', [3, 4, 5, 6])
Queue Overflow
Data removed:
array('i', [4, 5, 6])
First element is: 4
Data removed:
array('i', [5, 6])
---- QUEUE ----
array('i', [5, 6])

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```

b. Simulate a simple queuing system (e.g., customer service queue).

CODE :

```
import time

que = [("C1",2), ("C2",5), ("C3",7), ("C4",0)]

print(" ----- Customer Order ----- ")
print(que)

while que:
    Order, timer = que.pop(0)
    print(f" Serving: {Order} {timer} sec")
    time.sleep(timer)

print(que)

print("\nAbhishek S025")
```

OUTPUT :

```
----- Customer Order -----  
[('C1', 2), ('C2', 5), ('C3', 7), ('C4', 0)]  
Serving: C1 2 sec  
Serving: C2 5 sec  
Serving: C3 7 sec  
Serving: C4 0 sec  
[]
```

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Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?

3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?